



DATA SHEET

Vehicle-mounted Command & Control software for tactical operations

WINDOWS x64 Self-contained deployment	CoT / TAK Interoperable tactical data	NARROWBAND LEGIO v2 optimized transport	STANAG 4609 Video with KLV metadata
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MISSION PURPOSE

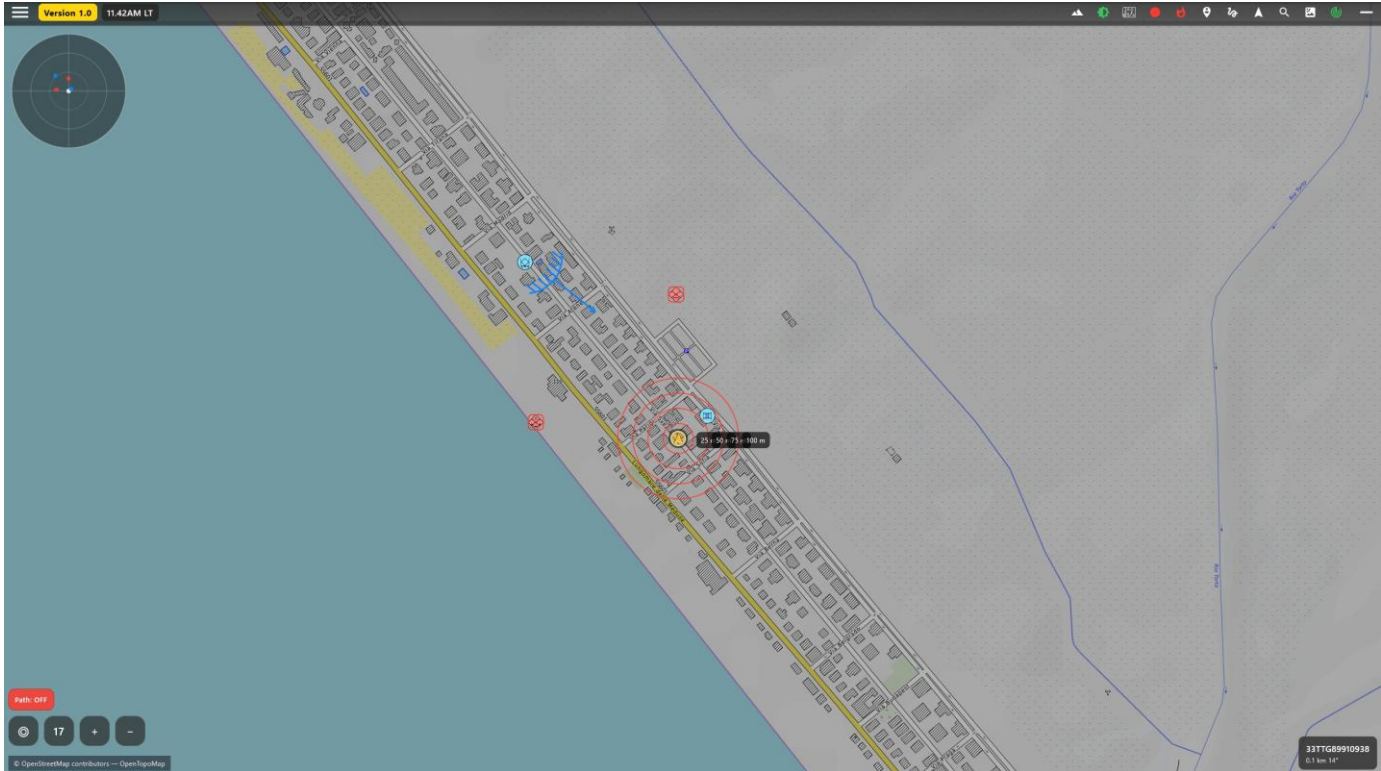
Praetorian C2 Vehicle provides a single vehicle-oriented operating environment for the Common Operational Picture, tactical messaging, CoT/TAK exchange, radio Situational Awareness, mission data, video feeds and terrain tools. The interface is designed for continuous map-centric operation and touchscreen use on Windows vehicle computers.

OPERATIONAL VALUE • Integrated COP and local/remote entity management • Reduced operator workload through a unified interface • Adaptable to conventional IP and constrained tactical links	MISSION ENVIRONMENT • Vehicle command posts and ruggedized Windows tablets • Tactical Ethernet, Wi-Fi and radio IP networks • Offline or connected operations with local mission data
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Product version 1.0 | Document edition 1.0 | 03 August 2026

01 OPERATIONAL CAPABILITIES

Map-centric functions for situational awareness, command, coordination and mission support.



Common Operational Picture with local vehicle position, tactical entities, range rings and map controls.

01 COP & TRACK MANAGEMENT

Map display, local and received entities, MGRS/LAT-LON, affiliation, history, distance, bearing and local dead reckoning.

02 TACTICAL MESSAGING

Presence, alerts, private chat, contacts, Data Packages and structured FORM MSG templates for operational reports.

03 MAPS & TERRAIN

Online and offline mapping, map package import, heatmap, DTED terrain data, slope profile and line-of-sight analysis.

04 DRAWING & PLANNING

Points, routes, areas, tactical graphics, free draw, patrol planning, path drawing, range fan and weapon ranges.

05 RADIO & SENSOR INPUT

Radio SA alias management, Harris CoT and NMEA track ingestion, internal GNSS via serial COM and camera integration.

06 VIDEO & MISSION DATA

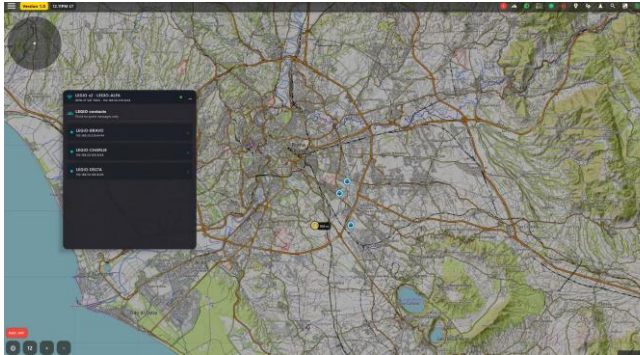
Live camera streaming, network video feeds, STANAG 4609/KLV display, logistics status and local mission repositories.

OPERATOR FOCUS

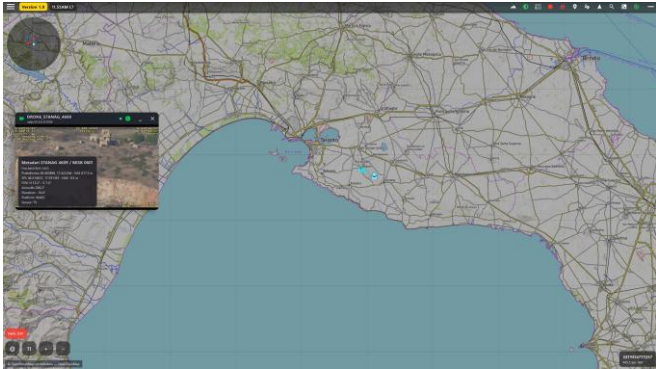
The map remains the primary operational workspace. Panels open over the COP so that network status, selected tracks and mission context remain visible during configuration and execution.

02 INTEROPERABILITY & TACTICAL NETWORKS

Multiple transport options support connected headquarters networks, peer-to-peer tactical links and radio-derived SA.



LEGIO v2 point-to-point workspace for constrained tactical links.



STANAG 4609 video with MISB 0601 KLV-derived platform and sensor data.

DOMAIN	INTERFACE / FORMAT	IMPLEMENTATION SCOPE
TAK ecosystem	Cursor on Target / TAK Server	Entity exchange, presence, chat, server connectivity and Data Package workflows with compatible TAK peers.
Tactical IP	UDP unicast / multicast	Configurable network inputs and outputs for low-latency tactical data and video transport.
Constrained links	LEGIO v2	Point-to-point Text, COP delta and Full COP modes with profiles for bandwidth, latency, jitter and MTU.
Tactical radios	Harris CoT; NMEA GLL/GGA/RMC	Radio track ingestion, source aliasing and publication of assigned tracks into the vehicle COP.
Video	STANAG 4609; MISB 0601 KLV	Network stream display plus platform position, sensor geometry and footprint when valid metadata are available.
Navigation	NMEA over serial COM	Internal GNSS acquisition from a configured serial device and network-derived position sources.

NARROWBAND DESIGN

LEGIO v2 allows the operator to select the transmission cost appropriate to the link: short text for minimum overhead, COP delta for incremental updates, or Full COP when a complete operational picture is explicitly required. Profile-driven parameters make the expected delay and payload constraints visible before transmission.

COMPATIBILITY	Actual interoperability depends on the configuration of the peer system, radio model and waveform, network routing, certificates, codecs and the completeness of incoming metadata.
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03 TECHNICAL SPECIFICATION

Implementation characteristics derived from the current Release 1.0 source tree and operator functions.

PARAMETRO	SPECIFICA
Product	Praetorian C2 Vehicle - Release 1.0
Target platform	Microsoft Windows 10 or later, 64-bit architecture
Application stack	Kotlin/JVM, JetBrains Compose Desktop, modular :core and :desktop project structure
Runtime deployment	Self-contained Windows package with Java 17, GStreamer 1.0 and required Microsoft Visual C++ runtime libraries
Local persistence	SQLite database plus application-owned files and preferences in the user profile
Map technologies	Online OSM-based view; offline map packages; OSM PBF import; TIFF/GeoTIFF image support
Terrain data	DTED import and use for slope and line-of-sight functions
Symbology	Military entity catalogue and MIL-STD-2525 tactical graphics rendering support
Networking	Configurable interfaces, UDP unicast/multicast inputs and outputs, TAK Server connectivity and LEGIO v2 peer links
GNSS / serial	NMEA position input through configurable Windows COM ports
Video	GStreamer-based camera and network video pipelines; UDP/MPEG-TS and STANAG 4609/KLV workflows
Mission data	Data Packages, local repositories, XLSX-based logistics and configuration workflows, PDF report generation
Operator interface	Map-centric desktop UI with touchscreen interaction and optional virtual keyboard
Data protection	Configurable crypto settings, complete Panic Zeroize and category-based Selective Zeroize

SECURITY, DEPLOYMENT & SUPPORT

CONTROLLED STORAGE Operational data are maintained in Praetorian-owned local paths. Offline maps, DTED, inbox data, graphics and settings can be managed independently.	ZEROIZE Panic Zeroize removes all local Praetorian data and resets credentials. Selective Zeroize erases only the authorized category.	PORTABLE INSTALLER The target computer does not require separate installation of Java, GStreamer or the Microsoft Visual C++ Redistributable.
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PROJECT RESOURCES

Website: moicanox.github.io/Praetorian-C2-Vehicle/ | **Support:** praetorian.c2@gmail.com
Installers, operator manuals, templates and release documentation are distributed through the project website and GitHub releases.